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Frosty

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1 Connics

1.1 Conic Sections and Conics

Problem 1

Determine the equation of each of the circles with the following centre and radius.

Solution. (a) Centre the origin, radius 1:

$$x^2 + y^2 = 1.$$

(b) Centre the origin, radius 4:

$$x^2 + y^2 = 16.$$

(c) Centre (3, 1), radius 2:

$$(x - 3)^2 + (y - 1)^2 = 4.$$

(d) Centre (3, 4), radius 3:

$$(x - 3)^2 + (y - 4)^2 = 9.$$

Problem 2

Determine the condition on the numbers f , g , and h in the equation

$$x^2 + y^2 + fx + gy + h = 0$$

for the circle to pass through the origin.

Solution. Clearly $(0, 0)$ must be a solution, so we require

$$0^2 + 0^2 + f(0) + g(0) + h = 0,$$

which gives $h = 0$. Thus the condition is $h = 0$.

Problem 3

Determine the centre and radius of the circles given by the following equations.

Solution. (a) $x^2 + y^2 - 2x - 6y + 1 = 0$. By Theorem 2 we have centre

$$\left(-\frac{1}{2}(-2), -\frac{1}{2}(-6)\right) = (1, 3)$$

and radius

$$\sqrt{\frac{1}{4}(-2)^2 + \frac{1}{4}(-6)^2 - 1} = \sqrt{1 + 9 - 1} = 3.$$

(b) $3x^2 + 3y^2 - 12x - 48y = 0$. Dividing through by 3 gives $x^2 + y^2 - 4x - 16y = 0$. By Theorem 2 we have centre

$$\left(-\frac{1}{2}(-4), -\frac{1}{2}(-16)\right) = (2, 8)$$

and radius

$$\sqrt{\frac{1}{4}(-4)^2 + \frac{1}{4}(-16)^2} = \sqrt{4 + 64} = \sqrt{68} = 2\sqrt{17}.$$

Problem 4

Determine the set of points (x, y) in $\mathbb{R}^2$ that satisfies each of the following equations.

Solution. (a) $x^2 + y^2 + x + y = 1$. By Theorem 2 we have centre $\left(-\frac{1}{2}, -\frac{1}{2}\right)$ and radius

$$\sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}}.$$

So the set is all points $(x, y) \in \mathbb{R}^2$ with distance $\sqrt{\frac{3}{2}}$ from $\left(-\frac{1}{2}, -\frac{1}{2}\right)$, i.e. a circle.

(b) $x^2 + y^2 - 2x + 4y + 5 = 0$. By Theorem 2 we have centre

$$\left(-\frac{1}{2}(-2), -\frac{1}{2}(4)\right) = (1, -2)$$

and radius

$$\sqrt{\frac{1}{4}(-2)^2 + \frac{1}{4}(4)^2 - 5} = \sqrt{1 + 4 - 5} = 0.$$

So the set is the single point $(1, -2)$.

(c) $2x^2 + 2y^2 + x - 3y - 5 = 0$. Dividing through by 2 gives

$$x^2 + y^2 + \frac{1}{2}x - \frac{3}{2}y - \frac{5}{2} = 0.$$

By Theorem 2 we have centre

$$\left(-\frac{1}{2}\left(\frac{1}{2}\right), -\frac{1}{2}\left(-\frac{3}{2}\right)\right) = \left(-\frac{1}{4}, \frac{3}{4}\right)$$

with radius

$$\sqrt{\frac{1}{4}\left(\frac{1}{2}\right)^2 + \frac{1}{4}\left(\frac{3}{2}\right)^2 + \frac{5}{2}} = \sqrt{\frac{1}{16} + \frac{9}{16} + \frac{40}{16}} = \sqrt{\frac{50}{16}} = \frac{5\sqrt{2}}{4}.$$

So the set is all points $(x, y) \in \mathbb{R}^2$ with distance $\frac{5\sqrt{2}}{4}$ from $\left(-\frac{1}{4}, \frac{3}{4}\right)$, i.e. a circle.

Problem 5

Determine which, if any, of the following pairs of intersecting circles are mutually orthogonal.

Solution. (a) $C_1 = \{(x, y) \mid x^2 + y^2 - 4x - 4y + 7 = 0\}$ and $C_2 = \{(x, y) \mid x^2 + y^2 + 2x - 8y + 5 = 0\}$. Notice

$$(-4)(2) + (-4)(-8) = -8 + 32 = 24, \quad 2(7 + 5) = 2(12) = 24.$$

So by Theorem 3, C_1 and C_2 are orthogonal.

(b) $C_1 = \{(x, y) \mid x^2 + y^2 + 3x - 6y + 5 = 0\}$ and $C_2 = \{(x, y) \mid 3x^2 + 3y^2 + 4x + y - 15 = 0\}$. Dividing through by 3 on C_2 gives

$$C_2 = \{(x, y) \mid x^2 + y^2 + \frac{4}{3}x + \frac{1}{3}y - 5\}.$$

Notice

$$(3)(\frac{4}{3}) + (-6)(\frac{1}{3}) = 4 - 2 = 2, \quad 2(5 + (-5)) = 2(0) = 0.$$

Since $2 \neq 0$, by Theorem 3, C_1 and C_2 are not orthogonal. ■

Problem 6

Find the equation of the line through the points of intersection of the circles

$$x^2 + y^2 - 3x + 4y - 1 = 0, \quad x^2 + y^2 + 5x - 6y + 3 = 0.$$

Solution. By Theorem 4 the equation of the line is the difference of the two circle equations:

$$(x^2 + y^2 - 3x + 4y - 1) - (x^2 + y^2 + 5x - 6y + 3) = 0,$$

which simplifies to

$$-8x + 10y - 4 = 0, \quad \text{i.e.} \quad 4x - 5y + 2 = 0. \quad \blacksquare$$

Problem 7

This question concerns the parabola E with equation $y^2 = x$ and parametric equations $x = t^2$, $y = t$ where $t \in \mathbb{R}$.

(a) Write down the focus, vertex, axis, and directrix of E .

Solution. Writing $y^2 = 4ax = x$, so $4a = 1 \implies a = \frac{1}{4}$. So it has focus $(\frac{1}{4}, 0)$ and directrix $x = -\frac{1}{4}$. Its vertex is $(0, 0)$ and its axis is the x -axis.

Let P and Q be coordinates on E so that

$$P = (t_1^2, t_1), \quad Q = (t_2^2, t_2).$$

If $t_1 \neq t_2$ then the chord PQ has slope

$$\frac{t_2 - t_1}{t_2^2 - t_1^2} = \frac{1}{t_2 + t_1}.$$

Since (t_1^2, t_1) lies on the line we have

$$(y - t_1) = \left(\frac{t_2 - t_1}{t_2^2 - t_1^2} \right) (x - t_1^2) \implies (y - t_1)(t_2^2 - t_1^2) = (t_2 - t_1)(x - t_1^2) \implies (y - t_1)(t_2 + t_1) = (x - t_1^2).$$

It passes through the focus $(\frac{1}{4}, 0)$ iff

$$(0 - t_1)(t_2 + t_1) = \left(\frac{1}{4} - t_1^2 \right) \implies -t_1 t_2 - t_1^2 = \frac{1}{4} - t_1^2 \implies -t_1 t_2 = \frac{1}{4} \implies t_1 t_2 = -\frac{1}{4}.$$

So the chord PQ is a focal chord precisely when $t_1 t_2 = -\frac{1}{4}$. ■